GÖKHAN "KHAN" BALIK, Ph.D.

Cybersecurity, AI & GIS Researcher | Systems Developer

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PROFESSIONAL SUMMARY

Cybersecurity, AI and GIS researcher and systems developer specializing in geospatial cyber risk intelligence, critical infrastructure resilience, and trustworthy AI-assisted decision support. Builds on a Ph.D. in GIS-based risk modeling, 14+ years of research (30 publications, 90 citations, 6 funded projects), and an M.S. in Cybersecurity (AI emphasis) in progress. Developer of four completed platforms spanning geospatial cyber risk intelligence (CyberGRIP), AI-assisted network discovery and security assessment, NIST-aligned risk assessment, and document-grounded RAG decision support. Hands-on experience in penetration testing, computer vision (YOLO, Mask R-CNN), and LLM/RAG systems.

CORE COMPETENCIES

Cybersecurity: Penetration Testing, Vulnerability Assessment, Network Security, Cyber Risk Assessment, NIST Frameworks, OWASP, AI Security

AI & Computer Vision: Large Language Models (LLMs), NLP, Retrieval-Augmented Generation (RAG), Explainable AI (XAI), YOLO, Mask R-CNN, PyTorch, TensorFlow, Scikit-learn

GIS & Critical Infrastructure: Geospatial Cyber Risk Intelligence, Critical Infrastructure Protection & Resilience, Infrastructure Interdependency & Network Analysis, Cascading Failure & Complex Systems Risk Modeling

EDUCATION

M.S. in Cybersecurity with an Emphasis in AI | Webster University, St. Louis, MO | Aug 2025 – May 2027 | GPA: 3.9 | Scholarship Recipient | Cybersecurity CTF Team Leader

A.S. in Computer Programming | Anadolu University, Turkey (Remote) | Sep 2024 – Feb 2027 | GPA: 3.4

Ph.D. in Landscape Architecture | Ege University, Turkey | Jul 2011 – Mar 2018 | Dissertation: Creating a GIS based Ecological Risk Assessment Model

M.S. in Landscape Architecture | Ege University, Turkey | Sep 2006 – Jan 2010

B.S. in Landscape Architecture | Ege University, Turkey | Sep 2001 – Aug 2006

PROFESSIONAL EXPERIENCE

Cybersecurity, AI & GIS Researcher and Systems Developer

Independent Research & Development, St. Louis, MO | Aug 2025 – Present

- Lead independent research and systems development at the intersection of cybersecurity, AI, GIS, computer vision, and critical infrastructure protection — four completed platforms detailed below.

Ethical Hacker, Programmer & IT Support Technician

StallionDC Ltd., U.K. (Remote, Part-Time) | Aug 2024 – Jul 2025

- Performed penetration testing and vulnerability assessments to identify security weaknesses, and documented mitigation guidance that strengthened data center security.
- Developed automation applications for Linux and network support workflows, improving operational efficiency by 25%.

Computer Vision Engineer (Internship)

Seattle-based Tech Company, Seattle, WA (Remote) | Oct 2023 – Feb 2024

- Developed a YOLO-based object detection model for satellite and UAV imagery, detecting small objects under variable atmospheric conditions.
- Built a real-time preprocessing pipeline for images, video, and live streams (normalization and quality enhancement), achieving over 90% precision.
- Developed two wind-turbine crack-detection workflows from UAV infrared and RGB imagery using YOLO and Mask R-CNN, supporting energy-infrastructure inspection.

Associate Professor, Researcher & Lecturer

Trakya University & Ege University, Turkey | Feb 2012 – Jul 2025

- Conducted 14+ years of funded interdisciplinary research in GIS, risk assessment, spatial analysis, and environmental modeling: 6 funded projects, 30 publications, 90+ citations, h-index 5, 1 other field award.
- Co-led an AI-based medical diagnostics project (publication in press); reviewer for two AI research projects funded by TUBITAK (Scientific and Technological Research Council of Turkey).
- Developed a student registration application that reduced processing time by 80%; organized national and international symposiums and workshops.

INDEPENDENT RESEARCH & ENGINEERING PROJECTS

CyberGRIP — *AI Geospatial CIKR Network Risk & Resilience Intelligence Platform*

- Built an AI-assisted platform integrating GIS, network science, and risk/resilience analytics to identify, analyze, and prioritize cyber risks across critical infrastructure (CIKR) systems.
- Modeled infrastructure interdependencies and cascading-risk pathways, applying deterministic risk scoring and grounded AI decision support for resilience planning, mitigation prioritization, and resource allocation.

Network Discovery & Security Assessment Tool — *AI-Assisted Asset Discovery and Topology Analysis*

- Developed a platform for automated network asset discovery, device classification, topology mapping, and dependency-aware cybersecurity assessment — the foundation for later work in cyber risk intelligence and critical infrastructure resilience.

AI-Assisted Risk Assessment Framework — *NIST-Aligned Cyber Risk Assessment*

- Developed a NIST-informed framework for cyber risk assessment, control-gap analysis, and mitigation prioritization, combining deterministic risk scoring with transparent, traceable AI-assisted decision support for organizational risk management.

Document-Grounded AI Assistant & RAG Framework — *Trustworthy, Evidence-Based Decision Support*

- Developed a retrieval-augmented generation (RAG) framework with evidence attribution and faithfulness verification, delivering explainable decision support for cybersecurity, risk analysis, and compliance workflows.

SELECTED PUBLICATIONS

- Balik, G. (2026). **Integrating GIS and Cybersecurity: Spatial Intelligence for Critical Infrastructure Risk and Resilience**. *GIS Odyssey Journal*, 6(1), 45–62. (Scopus-indexed)
- Balik, G., et al. (2026). **Privacy-Preserving Decision Support for Emergency Department Triage: Simulation-Based Benchmarking of On-Premises Retrieval-Augmented and Cloud-Based Large Language Models.** Submitted to an SCI-indexed Springer Nature journal; peer review completed.
- Malkoc, E., Kalaitzidis, C., & Balik, G. (2019). **The Role of Remote Sensing Technologies in Transboundary Protected Area Management.** *In Proceedings of the 15th International Conference "Standardization, Prototypes and Quality": A Means of Balkan Countries' Collaboration* (Edirne, Turkey, October 24–25, 2019). (Full-text conference proceeding)
- Balik, G., Aktas, E., Kalayci-Onac, A., & Birisci, T. (2017). **Vegetation Cover Change of Cesme Alacati Wind Power Plant Using Normalized Difference Vegetation Index (NDVI).** *Journal of International Environmental Application Science*, 12(3), 204–211.

TECHNICAL SKILLS

Security Tools: Burp Suite, Wireshark, Metasploit, Nmap, Kali Linux, Virtual Machines, Open-Source Tools

Development: Python, Bash, SQL, PHP, JavaScript, HTML/CSS, VBA, C#, Docker, Anaconda, GitHub

GIS & Remote Sensing: ArcGIS, AutoCAD, Spatial Analysis, Remote Sensing, Python Geospatial Libraries

CERTIFICATIONS

Zero to Mastery Academy: Complete Cybersecurity Bootcamp (2024); Complete Ethical Hacking Bootcamp (2025); Web Security & Bug Bounty — Penetration Testing (2025); Complete AI, Machine Learning & Data Science (2023); Complete Python Developer (2023); Statistics with Python (2024). TryHackMe courses and certificates (2025 – ongoing)